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CSA1407 - Compiler Design

Lexical Analysis Phase

Technical Presentation

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1. First Phase of Compilation

Front-End Foundation

Lexical Analysis is the **very first phase** of a compiler. It reads the source code character by character from left to right and converts it into a meaningful stream of atomic units called tokens.

Bridge to Syntax Analysis

It acts as an interface between the raw source code file and the **Parser (Syntax Analyzer)**, simplifying the grammar checking process in subsequent phases.

Symbol Table Interaction

During initial scanning, the Lexical Analyzer inserts identifiers (variable names, function names) into the Symbol Table to store attributes like name, type, and scope.



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2. Tokenization & Preprocessing



Tokenization

Converts input character sequences (lexemes) into standard numerical or symbolic pairs: .



Removes Whitespace

Strips unnecessary spaces, tabs, and newline characters from the source text, compressing the code for downstream processing.



Strips Comments

Filters out all single-line and multi-line comments so the parser processes only executable statements.



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3. Implementation via Finite Automata



Uses Finite Automata (FA): State transition systems (DFA & NFA) serve as the computational model to recognize valid token patterns.



Regular Expressions (RE): Token patterns (keywords, identifiers, numbers, operators) are formally specified using regular expressions.



Linear Time Execution: Deterministic Finite Automata (DFA) allow token recognition in single-pass linear time $O(n)$.



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4. Error Reporting & Key Takeaways

⚠ Lexical Error Detection

Identifies invalid characters, unclosed string literals, or malformed identifiers that fail to match any recognized pattern.

🛡 Error Recovery Strategies

Employs strategies like Panic Mode (deleting unexpected characters), inserting missing characters, or replacing illegal characters to continue scanning.

✅ Summary

The Lexical Analysis Phase acts as an efficient, automated scanner that cleans, structures, and transforms raw source code into standardized tokens using Finite Automata while detecting early input errors.